

Wazuh Installation Guide

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(This information is from official Wazuh documentation)

<https://documentation.wazuh.com/current/index.html>

Wazuh “Quickstart” Installation

System Requirements

Hardware Requirements

Hardware requirements are based on the number of endpoints or “agents”. Using the quickstart method of installation implies the Wazuh components (Indexer, Server, and Dashboard) are all on a single node. The quickstart method of installation can handle up to 100 protected endpoints.

Recommended hardware for quickstart deployment:

Agents	CPU	RAM	Storage (90 Days)
1-25	4 vCPU	8 GiB	50 GB
25-50	8 vCPU	8 GiB	100 GB
50-100	8 vCPU	8 GiB	200 GB

A distributed deployment (multi-node) is recommended for larger environments. Using a multi-node cluster configuration enables high availability and load balancing.

Operating System

The central components of Wazuh can be installed on a 64-bit Linux OS.

Recommended operating systems:

Amazon Linux 2	CentOS 7, 8
Red Hat Enterprise Linux	Ubuntu 16.04

Browser compatibility

The Wazuh dashboard is compatible with the following browsers:

Chrome 95 or later	Firefox 93 or later
Safari 13.7 or later	Other chromium-based browsers (variable)

Installing Wazuh

1. Download and run Wazuh installation assistant:

```
$ curl -sO https://packages.wazuh.com/4.8/wazuh-install.sh &&  
sudo bash ./wazuh-install.sh -a
```

Once finished, credentials for the Wazuh Dashboard will be displayed, confirming the successful installation.

```
INFO: --- Summary ---  
INFO: You can access the web interface  
https://<wazuh-dashboard-ip>  
      User: admin  
      Password: <ADMIN_PASSWORD>  
INFO: Installation finished.
```

2. Access Wazuh dashboard with 'https://<wazuh-dashboard-ip>/' and use credentials from the previous step.

When accessing the dashboard for the first time, there will be a warning message saying the certificate was not issued by a trusted authority. This is expected and the user has the option to accept the certificate as an exception or, alternatively, configure the system to use a certificate from a trusted authority.

NOTE: Passwords for Wazuh indexer and Wazuh API users can be found in the 'wazuh-passwords.txt' file inside 'wazuh-install-files.tar'. You can use the following command to view:
cutlTo uninstall Wazuh central components, run the Wazuh installation assistant from earlier and use either '-u' or '--uninstall'.

Wazuh Agents

From here, Wazuh agents can be deployed endpoints. Agents are lightweight and can be installed on clients, cloud instances, containers, and virtual machines. This will be shown later in the documentation.

Wazuh Indexer Assisted Installation

Requirements

Recommended Operating Systems

Amazon Linux 2	CentOS 7, 8
Red Hat Enterprise Linux	Ubuntu 16.04

Recommended Hardware

Component	Minimum		Recommended	
	RAM (GB)	CPU (cores)	RAM (GB)	CPU (cores)
Wazuh indexer	4	2	16	8

Disk Space

The amount of data depends on the generated alerts per second (APS). This table details the estimated disk space needed per agent to store 90 days of alerts on a Wazuh indexer server, depending on the type of monitored endpoints.

Monitored Endpoints	APS	Storage in indexer (GB/90 days)
Servers	0.25	3.7
Workstations	0.1	1.5
Network devices	0.5	7.4

For example, for an environment with 80 workstations, 10 servers, and 10 network devices, the storage needed on the Wazuh indexer server for 90 days of alerts is 230 GB.

Installing the Wazuh Indexer

The Wazuh indexer can be installed as a single-node or multi-node cluster using this method of installation.

Cluster Installation

The installation process is divided into three stages.

1. Initial configuration
2. Wazuh indexer nodes installation
3. Cluster initialization

Root user privileges are required to run the following commands.

1. Initial Configuration

Indicate your deployment configuration, create the SSL certificates to encrypt communications between the Wazuh components, and generate random passwords to secure your installation.

1. Download and install the Wazuh installation assistant and configuration file.

```
curl -sO https://packages.wazuh.com/4.8/wazuh-install.sh
curl -sO https://packages.wazuh.com/4.8/config.yml
```

2. Edit './config.yml' and replace the node names and IP values with the corresponding names and IP addresses. You need to do this for all Wazuh server, Wazuh indexer, and Wazuh dashboard nodes. Add as many node fields as needed.

```
nodes:
  # Wazuh indexer nodes
  indexer:
    - name: node-1
      ip: "<indexer-node-ip>"
    #- name: node-2
    # ip: "<indexer-node-ip>"
    #- name: node-3
    # ip: "<indexer-node-ip>"

  # Wazuh server nodes
  # If there is more than one Wazuh server
  # node, each one must have a node_type
  server:
    - name: wazuh-1
      ip: "<wazuh-manager-ip>"
      # node_type: master
    #- name: wazuh-2
    # ip: "<wazuh-manager-ip>"
    # node_type: worker
    #- name: wazuh-3
    # ip: "<wazuh-manager-ip>"
    # node_type: worker

  # Wazuh dashboard nodes
  dashboard:
    - name: dashboard
      ip: "<dashboard-node-ip>"
```

3. Run the Wazuh installation assistant with the option ‘--generate-config-files’ to generate the Wazuh cluster key, certificates, and passwords necessary for the installation. These files can be found in ‘./wazuh-install-files.tar’.

```
bash wazuh-install.sh --generate-config-files
```

4. Copy the ‘wazuh-install-files.tar’ file to all servers of the distributed deployment including the Wazuh server, indexer, and dashboard nodes. Can be done using the ‘scp’ utility.

2. Wazuh Indexer nodes installation

Install and configure the Wazuh indexer nodes.

1. Download the Wazuh installation assistant.

```
curl -sO https://packages.wazuh.com/4.8/wazuh-install.sh
```

2. Run the Wazuh installation assistant with the option ‘--wazuh-indexer’ and the node name to install and configure the Wazuh indexer. The node name must be the same one that was used in ‘config.yml’ for the initial configuration.

NOTE: Ensure that a copy of ‘wazuh-install-files.tar’, created during the initial configuration, is placed in your working directory.

```
bash wazuh-install.sh --wazuh-indexer node-1
```

Repeat this step for every Wazuh indexer node in your cluster. Then continue with initializing your single or multi-node cluster in the next stage.

3. Cluster initialization

The final stage of installing the Wazuh indexer single-node or multi-node cluster consists of running the security admin script.

1. Run the Wazuh installation assistant with option '--start-cluster' on any Wazuh indexer node to load the new certificates info and start the cluster.

```
bash wazuh-install.sh --start-cluster
```

NOTE: The cluster only needs to be initialized once, there is no need to run this command on every node.

Testing the Cluster Installation

1. Run to get the admin password:

```
tar -axf wazuh-install-files.tar
wazuh-install-files/wazuh-passwords.txt -O | grep -P "'admin'"
-A 1
```

2. Run the following command to confirm that the installation is successful. Replace '<ADMIN_PASSWORD>' with output from the previous step and replace '<WAZUH_INDEXER_IP>' with Wazuh indexer IP address.

```
curl -k -u admin:<ADMIN_PASSWORD> https://<WAZUH_INDEXER_IP>:9200
```

3. Run the following to confirm the cluster is working. Again, replace '<WAZUH_INDEXER_IP>' and '<ADMIN_PASSWORD>'.

```
curl -k -u admin:<ADMIN_PASSWORD>
https://<WAZUH_INDEXER_IP>:9200/_cat/nodes?v
```

Next

With the Wazuh indexer installed, proceed to installing the Wazuh server.

Wazuh Server Assisted Installation

Installing the Wazuh server

Install the Wazuh server as a single-node or multi-node cluster using the assisted installation method. The Wazuh server analyzes the data received from the agents triggering alerts when it detects threats and anomalies. This central component includes the Wazuh manager and Filebeat.

Wazuh Server Cluster Installation

1. Download Wazuh installation assistant

```
curl -sO https://packages.wazuh.com/4.8/wazuh-install.sh
```

2. Run installation assistant with the option '--wazuh-server' followed by the node name to install the Wazuh server. This node name must be the same as the one configured in 'config.yml' from the initial configuration.

NOTE: Ensure a copy of 'wazuh-install-files.tar', created during the initial configuration is in your working directory.

```
bash wazuh-install.sh --wazuh-server wazuh-1
```

The Wazuh server is now installed.

- If you want Wazuh server as a single node-cluster, you can proceed to installing the Wazuh dashboard.
- If you want Wazuh server as a multi-node cluster, this must be repeated on every Wazuh server node.

Next

Proceed with installing the Wazuh dashboard.

Wazuh Dashboard Assisted Installation

Install and configure the Wazuh dashboard using the assisted installation method. Wazuh dashboard is a flexible and intuitive web interface for mining and visualizing security events and archives.

Installing the Wazuh Dashboard

1. Download the Wazuh installation assistant. You can skip this step if you have already installed Wazuh indexer on the same server.

```
curl -sO https://packages.wazuh.com/4.8/wazuh-install.sh
```

2. Run the Wazuh installation assistant with the option '--wazuh-dashboard' and the node name to install and configure the Wazuh dashboard. The node name must be the same one used in '.config.yml' for the initial configuration, for example, 'dashboard'.

NOTE: Ensure a copy of the 'wazuh-install-files.tar' file, created during the initial configuration step, is placed in your working directory.

```
bash wazuh-install.sh --wazuh-dashboard dashboard
```

The default Wazuh web user interface port is 443, used by the Wazuh dashboard. You can change this port using the optional parameter '-pl -port <port_number>'. Some recommended ports: 8443, 8444, 8080, 8888, 9000.

Once the Wazuh installation is completed, the output shows the access credentials and a message that confirms that the installation was successful.

```
INFO: --- Summary ---
INFO: You can access the web interface https://<wazuh-dashboard-ip>
      User: admin
      Password: <ADMIN_PASSWORD>

INFO: Installation finished.
```

You now have installed and configured Wazuh. Find all passwords that the Wazuh installation assistant generated in the 'wazuh-passwords.txt' file inside the 'wazuh-install-files.tar' archive. To print, run the following:

```
tar -O -xvf wazuh-install-files.tar
wazuh-install-files/wazuh-passwords.txt
```

3. Access the Wazuh web interface.

- URL: <https://<wazuh-dashboard-ip>>
- Username: admin
- Password: <ADMIN_PASSWORD>

When you access the Wazuh dashboard for the first time, the browser shows a warning message stating that the certificate was not issued by a trusted authority. An exception can be added in the advanced options of the web browser. For increased security, the 'root-ca.pem' file previously generated can be imported to the certificate manager of the browser instead. Alternatively, a certificate from a trusted authority can be configured.

Next

The Wazuh environment is now ready, and you can proceed with installing the Wazuh agent on the endpoints to be monitored. Wazuh agents can now be installed on endpoints.

Wazuh Agent

The Wazuh agent is multi-platform and runs on the endpoints that the user wants to monitor. It communicates with the Wazuh server, sending data in near real-time through an encrypted and authenticated channel.

The agent was developed considering the need to monitor a wide variety of different endpoints without impacting their performance. It is supported on the most popular operating systems, and it requires 35 MB of RAM on average.

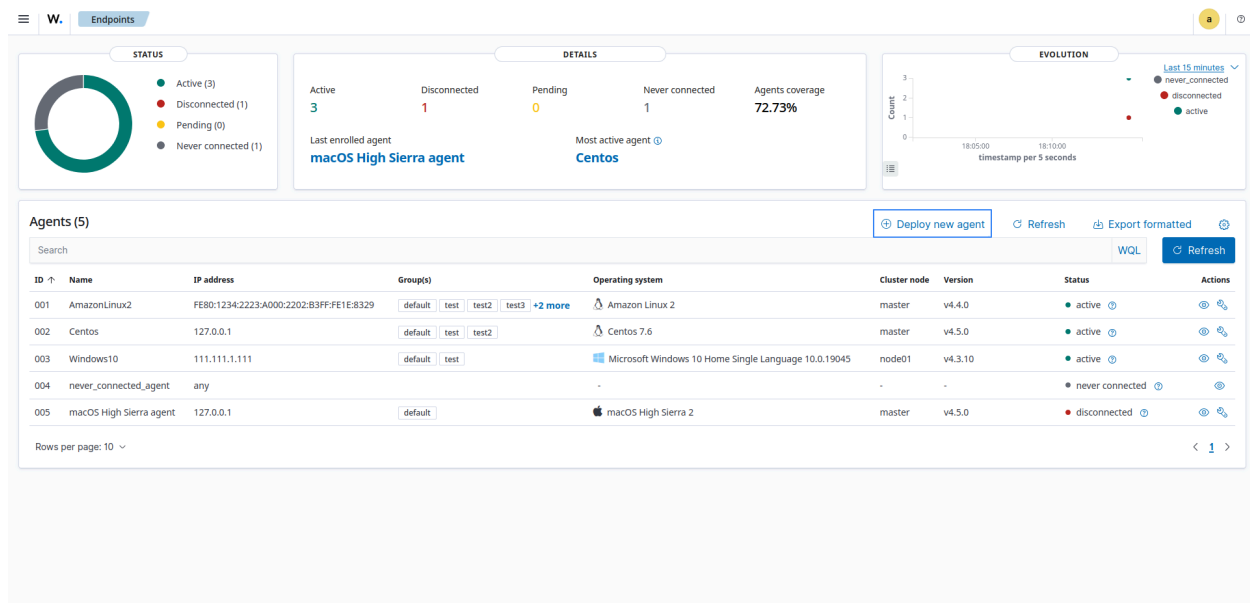
The following links are installation steps for the Wazuh Agents on different operating

systems:

- [Linux](#)
- [Windows](#)
- [macOS](#)
- [Solaris](#)
- [AIX](#)
- [HP-UX](#)

If you are deploying Wazuh in a large environment, with a high number of servers or endpoints, keep in mind that this deployment might be easier using automation tools such as Puppet, Chef, SCCM, or Ansible.

You can also deploy a new agent following the instructions in the Wazuh dashboard. Go to **Endpoints Summary**, and click on **Deploy new agent**.



Then the Wazuh dashboard will show you the steps to deploy a new agent.

× Close

Deploy new agent

Select the package to download and install on your system:



LINUX

- ☐ RPM amd64 ☐ RPM aarch64
☐ DEB amd64 ☐ DEB aarch64



WINDOWS

- ☒ MSI 32/64 bits



macOS

- ☐ Intel
☐ Apple silicon

④ For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ④

10.0.0.198

☐ Remember server address

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: ④

Windows

④ The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups: ④

default ×



4 Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.8.0-1.msi -OutFile $(env.tmp)\wazuh-agent; msixec.exe /i $(env.tmp)\wazuh-agent /q WAZUH_MANAGER='10.0.0.198' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='Windows'
```

④ Requirements

- You will need administrator privileges to perform this installation.
- PowerShell 3.0 or greater is required.

Keep in mind you need to run this command in a Windows PowerShell terminal.

5 Start the agent:

```
NET START WazuhSvc
```